

Akarsh Vijjapu

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PROFESSIONAL SUMMARY

Backend-focused Full Stack Developer with hands-on experience building REST APIs, real-time WebSocket systems, and AI-integrated applications using Python, FastAPI, SQLite, and SQLAlchemy ORM. Delivered three end-to-end projects featuring full CRUD operations, database integration, JWT authentication, and role-based access control. Pursuing B.Tech in CSE (CGPA: 8.86) with 100+ DSA problems solved on LeetCode and HackerRank.

TECHNICAL SKILLS

Languages: Python, Java

Database: SQLite, SQLAlchemy ORM, SQL

Tools: Git, GitHub, VS Code

Backend: FastAPI, REST API Design, WebSockets, CRUD Operations, Background Tasks

Frontend: HTML5, CSS3, JavaScript

Core CS: Data Structures, Algorithms, OOP

PROJECTS

MediQueue — Hospital Queue Management System

github.com/akarsh277/MediQueue

Python · FastAPI · SQLite · SQLAlchemy ORM · REST APIs · JWT Auth · RBAC

- Architected a queue management backend with full CRUD operations for patient records, appointments, and token assignments using SQLite and SQLAlchemy ORM.
- Implemented JWT-based authentication and role-based access control (RBAC) across Admin, Doctor, and Patient modules with strict permission enforcement per role.
- Added rate limiting on critical API endpoints and background tasks for automated queue state updates and notifications.

CureBot — AI-Powered Medical Assistant

github.com/akarsh277/CureBot

Python · FastAPI · REST APIs · LLM API Integration

- Built an AI symptom assessment chatbot backend integrating an LLM API for automated symptom responses and health suggestions.
- Implemented response caching to reduce redundant AI API calls and structured background tasks for async AI response logging without blocking the main request thread.

DisasterLink — Real-Time Emergency Operations Platform

github.com/akarsh277/Disaster

Python · FastAPI · SQLite · SQLAlchemy ORM · WebSockets · JWT Auth · REST APIs

- Engineered a real-time disaster coordination platform with persistent WebSocket connections and broadcast logic pushing live alerts to citizens, volunteers, and authorities simultaneously.
- Implemented JWT-based authentication for volunteer and authority login; applied rate limiting on report submission endpoints to prevent spam during high-traffic events.
- Built event-driven backend with background tasks for alert escalation and automated status updates without polling overhead.

EDUCATION

B.Tech — Computer Science Engineering — Pragati Engineering College, Surampalem

2024–2028 · CGPA: 8.86

Intermediate — MPC — Narayana Junior College, Kakinada

2022–2024 · 94%

SSC — Sama School, Jagannaickpur

2021–2022 · 96%

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL — Programming in Java (core Java and OOP concepts)
- Scaler Topics — Python Course for Beginners: Mastering the Essentials (Aug 2025)
- Be10x — AI Tools & ChatGPT Workshop (Feb 2026)
- Brain O Vision — Web App Development Workshop & 24Hr Hackathon, Pragati Engineering College (Feb 2025)
- Participant — MindSprint 2K25 Hackathon (backend development under competitive, time-constrained conditions)
- Participant — Srujana Hackathon (team-based solution for a real-world engineering problem)
- Solved 100+ algorithmic problems on LeetCode and HackerRank — arrays, trees, graphs, dynamic programming